Enrollment No. 039

MID-TERMEXAMINATION

(CBCS)

Course Name:B. Tech (Mechanical and Automation Engineering), Semester:3rd

(21st Oct, 2022)

Subject Code:BMA-203 Subject: Strength of Materials

Time : 1 ½ Hours Maximum Marks : 30

Note:Q. 1 is compulsory. Attempt any two questions from the rest.

Q1 (2.5*4)

(a) What is stress? Explain with the help of suitable diagram.

(b) What is point of contraflexure? Explain with a suitable diagram.

(c) Define factor of safety. What is the significance of factor of safety in designing of components/elements.

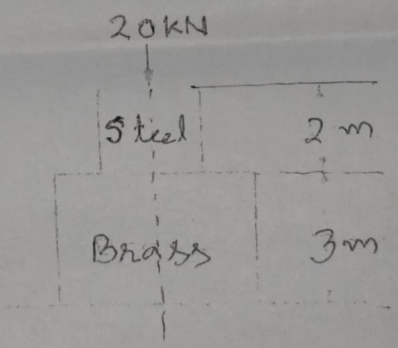
(d) Define the term Poison's ratio.

Q2 (10)

Draw and explain in detail the various salient points of the stress-strain curve for mild steel.

Q3 (5,5)

(a) A composite bar of length 5m is made of two materials as shown in the figure below. The first 3m of its length is made of brass and is of 7.5 cm^2 in cross-section and the remainder of its length is of steel and is of 5 cm^2 in cross-section. Determine the total compression of the bar under a load of 20 kN. $E_{Steel} = 210GN/m^2$, $E_{brass} = 84GN/m^2$.



(b) The principal stresses at a point across two perpendicular planes are 75 MN/m^2 (tensile) and 35 MN/m^2 (tensile). Find the normal, tangential stresses and the resultant stress and its obliquity on a plane at 20° with the major principal plane.

Q4 (10)

Draw the shear force and bending moment diagrams for the beams loaded as shown in the figure given below. Clearly mark the position of maximum bending moment and determine its value.

